Chapter 15, Problem 27P

Problem

Determine the inverse Laplace transform of each of the following functions:

(a) $F(s) = \frac{1}{s} + \frac{2}{s+1}$

(b) $G(s) = \frac{3s+1}{s+4}$

(c) $H(s) = \frac{4}{(s+1)(s+3)}$

(d) $J(s) = \frac{12}{(s+2)^2(s+4)}$

Step-by-step solution

Step 1 of 6

(a)

Consider the following expression.

$$F(s) = \frac{1}{s} + \frac{2}{s+1}$$

Use the following inverse Laplace expressions.

$$\frac{1}{s} \Leftrightarrow u(t)$$

$$\frac{1}{s+a} \Leftrightarrow e^{-at}u(t)$$

Apply inverse Laplace transform to $F(s)$.

$$f(t) = u(t) + 2e^{-t}u(t)$$

Therefore, the inverse Laplace transform of $F(s)$ is $\boxed{u(t) + 2e^{-t}u(t)}$.

Step 2 of 6

(b)

Consider the following expression.

$$\begin{aligned} G(s) &= \frac{3s+1}{s+4} \\ &= \frac{3(s+4-4)+1}{s+4} \\ &= \frac{3(s+4)-12+1}{s+4} \\ &= \frac{3(s+4)}{s+4} - \frac{11}{s+4} \\ &= 3 - \frac{11}{s+4} \end{aligned}$$

Use the following inverse Laplace expressions.

$$\frac{1}{s+a} \Leftrightarrow e^{-at}u(t)$$
$$1 \Leftrightarrow \delta(t)$$

Apply inverse Laplace transform to $G(s)$.

$$g(t) = 3\delta(t) - 11e^{-4t}u(t)$$

Therefore, the inverse Laplace transform of $G(s)$ is $\boxed{3\delta(t) - 11e^{-4t}u(t)}$.

Step 3 of 6

(c)

Consider the following expression.

$$H(s) = \frac{4}{(s+1)(s+3)}$$

Use partial fraction expansions.

$$\frac{4}{(s+3)(s+1)} = \frac{A}{s+1} + \frac{B}{s+3}$$

$$4 = A(s+3) + B(s+1) \quad \ldots\ldots (1)$$

Determine the value of *A*.

Substitute $s = -1$ in equation (1).

$$4 = A(-1+3) + B(-1+1)$$
$$4 = A(2) + 0$$
$$A = 2$$

Determine the value of *B*.

Substitute $s = -3$ in equation (1).

$$4 = A(-3+3) + B(-3+1)$$
$$4 = 0 + B(-2)$$
$$B = -2$$

Step 4 of 6

Write the expression of $H(s)$.

$$H(s)=\frac{2}{s+1}-\frac{2}{s+3}$$

Apply inverse Laplace transform on both sides.

$$h(t)=2e^{-t}u(t)-2e^{-3t}u(t)$$

Therefore, the inverse Laplace transform of $H(s)$ is $\boxed{2e^{-t}u(t)-2e^{-3t}u(t)}$.

Step 5 of 6

(d)

Consider the following expression.

$$J(s)=\frac{12}{(s+2)^2(s+4)}$$

Use partial fraction expansions.

$$\frac{12}{(s+2)^2(s+4)}=\frac{A}{s+2}+\frac{B}{(s+2)^2}+\frac{C}{(s+4)}$$

$$12=A(s+2)(s+4)+B(s+4)+C(s+2)^2 \quad \ldots\ldots (2)$$

Determine the value of *B*.

Substitute $s=-2$ in equation (2).

$$12=A(-2+2)(s+4)+B(-2+4)+C(-2+2)^2$$
$$12=0+B(2)+0$$
$$B=6$$

Determine the value of *C*.

Substitute $s=-4$ in equation (2).

$$12=A(-4+2)(-4+4)+B(-4+4)+C(-4+2)^2$$
$$12=0+0+C(-4+2)^2$$
$$C=3$$

Compare coefficients of ‘s’ in equation (2).

$$0=6A+B+4C$$
$$0=6A+6+4(3)$$
$$A=-3$$

Step 6 of 6

Write the expression of $J(s)$.

$$J(s) = \frac{-3}{s+2} + \frac{6}{(s+2)^2} + \frac{3}{(s+4)}$$

Use the following inverse Laplace expressions.

$$\frac{1}{s+a} \Leftrightarrow e^{-at}u(t)$$
$$\frac{1}{(s+a)^2} \Leftrightarrow te^{-at}u(t)$$

Apply inverse Laplace transform on both sides.

$$j(t) = -3e^{-2t}u(t) + 6te^{-2t}u(t) + 3e^{-4t}u(t)$$

Therefore, the inverse Laplace transform of $J(s)$ is $\boxed{-3e^{-2t}u(t) + 6te^{-2t}u(t) + 3e^{-4t}u(t)}$.